PHD13 Public finance: Final assignment

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1 Setting

Solving the model in Julia

The assignment is completed in Julia, offering several improvements to the Matlab counterpart.

- The Julia implementation is highly optimised, solving the full model in under 0.5 seconds and making it an excellent framework for computationally heavy structural estimation.
- The model relies on Young's (2010) non-stochastic method to simulate the cross-sectional distribution, effectively eliminating the finite-sample noise and errors inherent in standard Monte Carlo simulations.
- Monte Carlo simulations are still produced, to be used in regressions or more detailed life-cycle analysis.
- To maximise readability, the code uses Unicode characters that map directly to the formal mathematical notation used in the theoretical exposition.

The complete codebase can be found and pulled from my GitHub repository, with all necessary setup instructions detailed in the project's README.

The model is set up to allow for solving it across multiple settings.

1. **Without extensive margin labour supply**: This mimics the code used for previous assignments.
2. **With extensive margin labour supply**: This features the endogenous extensive labour supply choice.

I solve the partial equilibrium model, where a household makes consumption-saving and labour participation decisions leading to the following value function:

$$V_t\left(a_t, y_t\right) = \max_{c_t, a_{t+1}, h_t} \left\{ \frac{c_t^{1-\gamma}-1}{1-\gamma} + \chi \frac{(L-h_t)^{1-\sigma}-1}{1-\sigma} + \beta\, \mathbb{E}_t\left[V_{t+1}\left(a_{t+1}, y_{t+1}\right) \mid y_t\right] \right\}. \tag{1}$$

The model is solved in a two-step procedure.

1. **Backward induction**: Leveraging that $V_{T+1} = 0$, I solve the model backwards. In `fnLastPeriod!`, I initialise the last period's decisions, when $a_{T+1}\left(a_t, y_t\right) = 0 \;\forall\; a_T, y_T$. This allows for computing $c_T\left(a_t, y_t\right)$ and $V_T\left(a_T, y_T\right)$. Then, for $t \in (T-1, T-2, \ldots, 2, 1)$ and for each $\left(a_t, y_t\right) \in \mathscr{A} \times \mathscr{Y}$:
   (a) Compute $\mathbb{E}_{t-1}\left[V_t\left(a_t, y_t\right) \mid y_{t-1}\right]$ (in `fnBackwardInduction!`).
   (b) Find the optimal asset policy, accounting for the minimum allowed consumption and borrowing constraint (in `fnFindAssets`). This is done for both working and non-working households.
   (c) Compare the values for working and non-working, which allows for determining the labour supply policy.
   (d) Compute the value function, $V_t\left(a_t, y_t\right)$

2. **Forward simulation**: Assume that all households enter with no assets and that the initial income shock is drawn from its steady state distribution. For each period $t \in \{2, 3, \ldots, T\}$, compute the mass of workers transitioning to the specific (a_{t+1}, y_{t+1}) grid. Because the chosen savings level a_{t+1} almost never lands exactly on a discrete grid point, the code splits the probability mass. It assigns a weighted fraction of the agents to the grid point just below their target, and the rest to the grid point just above it.

In Figure 1, I present the key statistics based on the model solved using the above procedure and the MC simulation. The main differences arise in the median wealth and consumption statistics, in Figures 1a and 1b respectively.

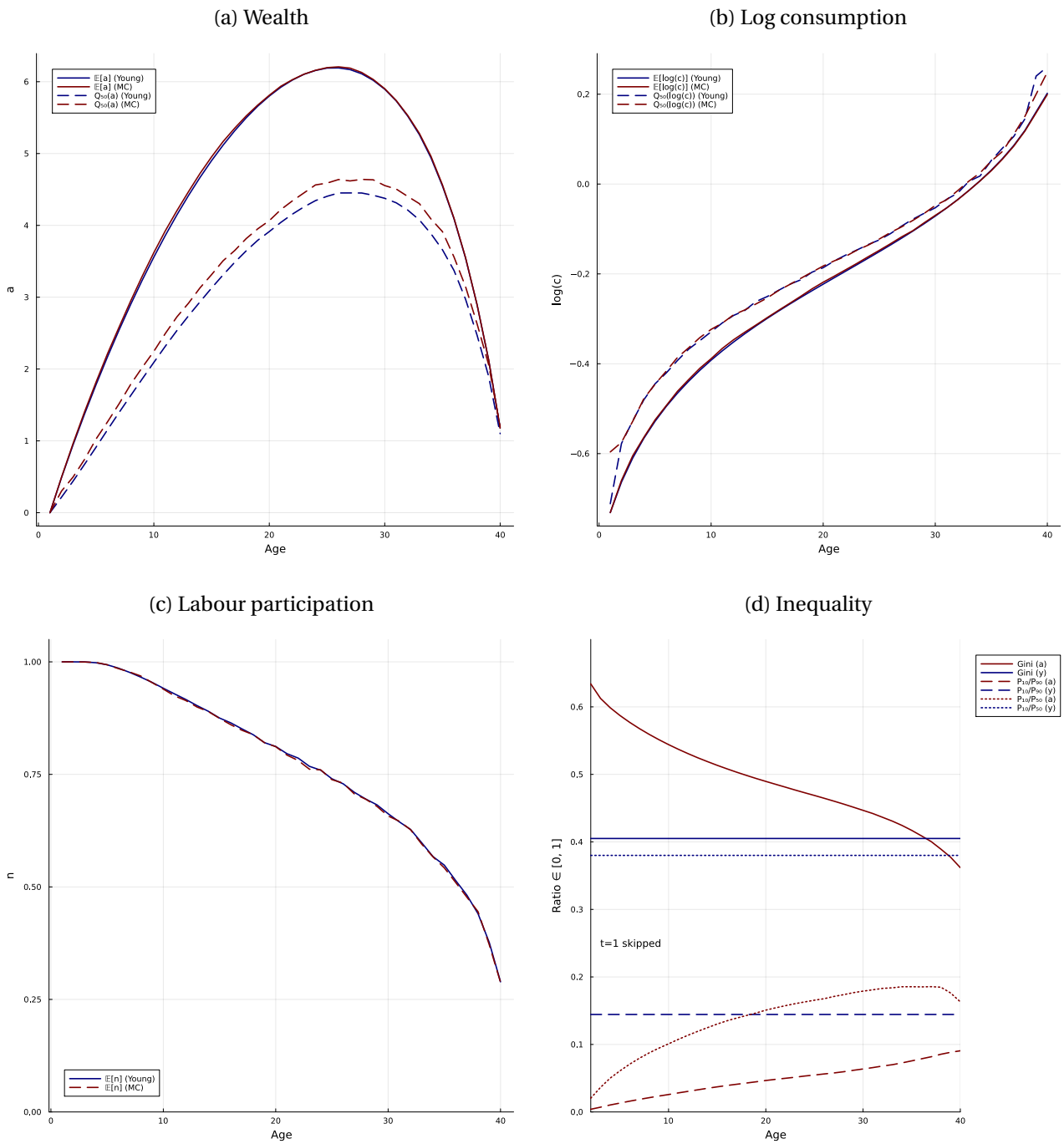


Figure 1: Life-cycle profiles.

2 Estimation strategy

While the exercise asks to provide a set of parameters that produces a non-trivial labour supply dynamics, as evidenced in Figure 2c, I provide further exposition on how I would do it in a proper way. It aims to show what I would do in the "best world." Table 1 summarises my approach and parameter values used.

1. The income and asset grid parameters would either be manually set (e.g., the number of grid points) or matched outside the model to the earnings data. In this version of the model, I also normalise the average income to 1.

2. Building on the best practices of macroeconomists, I would "borrow" certain parameter values from the literature, e.g.,$\beta = 0.99$ and $r = \frac{1}{\beta} - 1$.

3. γ would ideally be estimated from the consumption profiles in the (synthetic) data. This is precisely what I attempt to do in `analysis/EstimationGamma.jl`, but the exercise fails due to the binary nature of labour supply. As before, I pick a conventional value of 2 for γ.

   - Another way of "saving" the econometric estimation is to make income depend on a set of demographic characteristics, $\boldsymbol{Z}_t$, which this exercise abstracts from doing.

4. The last parameters left are χ and σ. With their collinearity, they cannot be estimated at the same time, at least with this synthetic data. I set $\sigma = 0.5$, based on what I would expect to see in the literature.

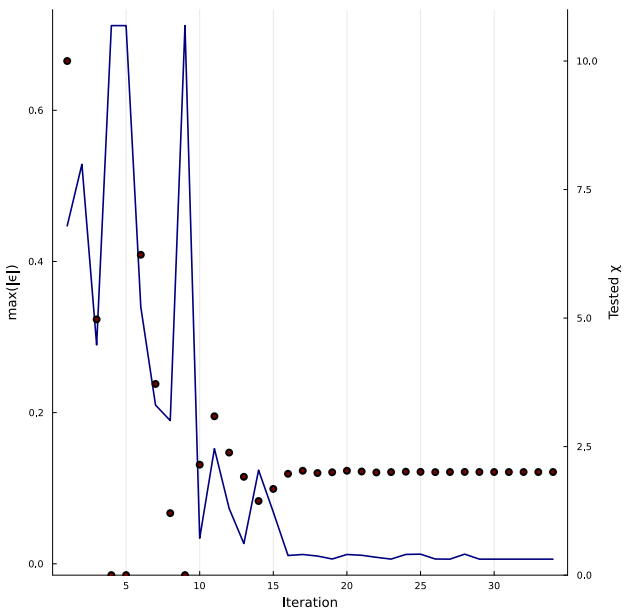
Parameter	**Value**	**Identification method**
A. Structural parameters		
Consumption utility parameter (γ)	2.0	Estimation / Literature
Disutility of labour shifter (χ)	2.0	Simulated method of moments
Labour utility parameter (σ)	0.5	Literature
Discount factor (β)	0.99	Literature
Leisure endowment (L)	1.0	Manual choice
Hours if working (h)	1/3	Manual choice
Interest rate (r)	$1/\beta - 1$	Externally matched
B. Lifecycle parameters		
Years in the labour market (T)	40	Manual choice
Consumption floor ($\underline{c}$)	10^{-8}	Manual choice
Simulated workers (S)	10,000	Manual choice
C. Income grid (Rouwenhorst method used, normalised $\mathbb{E}[y] = 1$)		
Persistence (ρ)	0.75	Externally matched
Shock standard deviation (σ^y)	0.50	Externally matched
Income grid size (N^y)	40	Manual choice
D. Assets grid (non-linear)		
Assets grid size (N^a)	80	Manual choice
Minimum assets ($\underline{a}$)	0.0	Manual choice
Maximum assets ($\overline{a}$)	80.0	Manual choice

Note: The income process parameters (ρ, σ^y) can be estimated externally using the earnings data (I pick relatively plausible values). The risk aversion parameter (γ) can be estimated via the Euler equation using lagged cohort variance as an instrument, albeit the current model formulation is not conducive of doing so (I pick a standard macroeconomic literature value). The labour disutility shifter (χ) is calibrated via Simulated Method of Moments targeting the lifecycle employment profile.

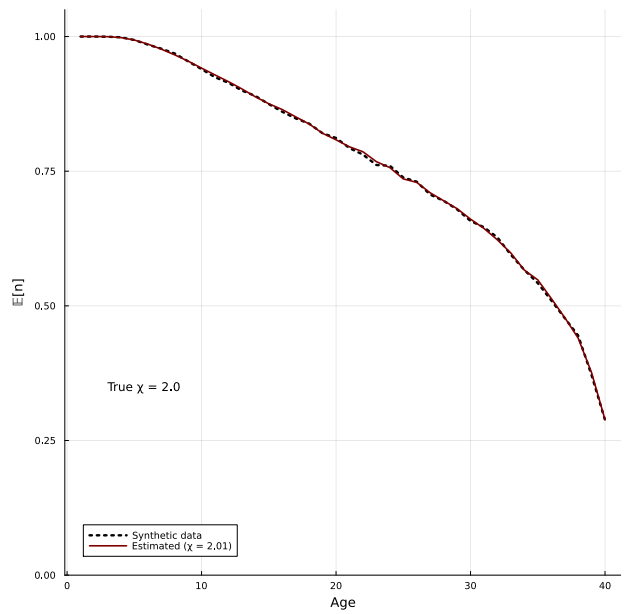
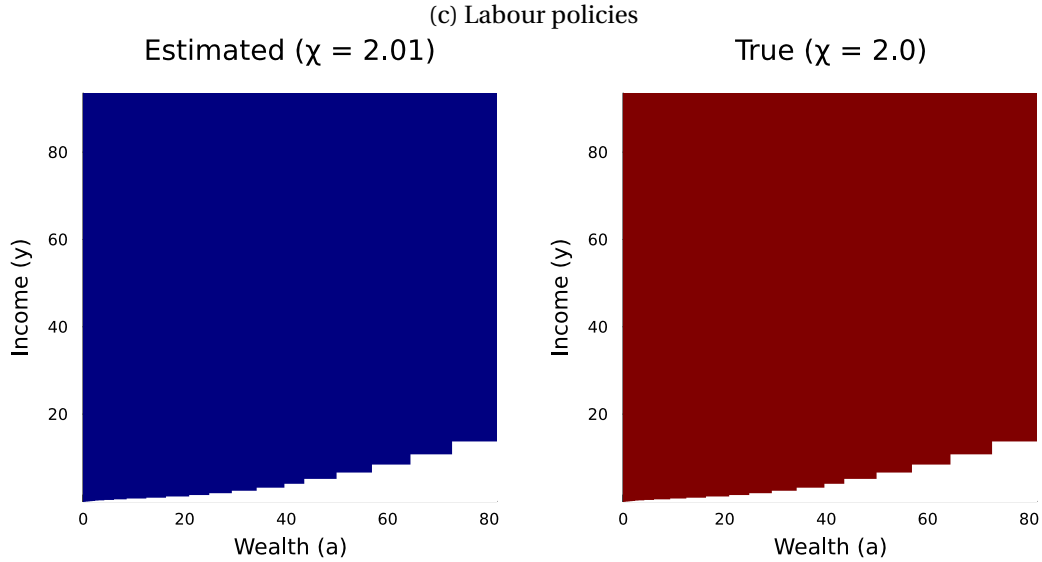
Table 1: Model parameters and estimation strategy.

Figure 2a summarises the calibration's relatively fast convergence (though, it's only one parameter). Julia takes around 14 iterations to arrive at the near-perfect value of χ, wrapping the calibration in under 25 seconds. Figure 2b features the model's fit as compared with the synthetic data. Given that I build the model to match the exact labour participation profile, it's expected that it's near perfectly matched. Figure 1 showcases the policies under the estimated and true versions.

(a) Convergence statistics



(b) Model fit



(c) Labour policies